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Integrated Pulsed Neutron Logging and Nuclear Modelling for Formation Saturation and Gravel Pack Integrity Evaluation in Complex Sand Control Completions

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Abstract

Managing both sand and water production remains a fundamental challenge in brownfield reservoir management. Wire Wrapped Screen (WWS) completions are widely adopted as part of the sand control strategies. However, the complex near-wellbore environment in such completions often masks true formation signals in behind-casing evaluation. This study aims to demonstrate an integrated approach using the Multi Detector Pulsed Neutron (MDPN) tool, pre-job Monte-Carlo N-Particle (MCNP) nuclear modelling and advanced interpretation workflow for both formation saturation and gravel pack integrity evaluation.

Over 10 wells across Asia were accessed, with two detailed case studies presented. MDPN Gas and Carbon/Oxygen (C/O) modes were deployed in cased well with double or single tubing across WWS Gravel Pack (GP) section. The MCNP nuclear modelling simulated tool responses for expected saturation envelope, incorporating effects from changes in fluid type from all tubing-screen-casing interfaces and known reservoir properties. This workflow leveraged the deeper depth of investigation (DOI) of Far and Long detectors against the shallow DOI from Proximal and Near detectors to separate the near-wellbore sand screen response from the true formation signals. In Gas mode, time-spectrum inelastic and capture ratio measurements, combined with detailed GP-specific MCNP models, identified whether the gas responses were originated from trapped annular gas below the unperforated blank pipe sections, from the formation, or from both. MCNP modeling attributes were adjusted to account for trapped gas effects on saturation envelopes.

Results from one case study indicated significant formation gas in the unperforated interest zone and later post-logging perforation confirmed gas flow. In another, C/O dual-saturation results using different detector ratio mixtures revealed probable water breakthroughs in existing perforation zones, aligned with observation of high water cut. Gas cap expansion could have occurred due to reservoir depletion. GP integrity evaluation was completed by adopting silicon activation analysis using additional bottom Gamma Ray (GR) sensor measurements, allowing determination of gravel top, and packing percentage. This enabled identification of localized GP deterioration within the WWS interval, providing critical input for operators on completion remediation decisions when required.

The integrated MDPN-MCNP approach proved effective in complex sand control completions, delivering accurate behind-casing saturation answers and reliable GP condition assessment without costly intervention. Detector ratio mixture flexibility from MDPN tool has gained user's confidence in distinguishing true reservoir signals even in challenging GP completions. This capability supports improved production optimization and long-term sustainability in mature assets.

Introduction

Sand production occurs when production-induced forces exceed the restraining strength of formation grains, causing issues with flow line and surface production equipment. The primary challenge is sanding, where loosely bound sand grains are mobilized and produced along with hydrocarbons. This leads to not only a cease in production but also incur additional costs for sand remediation or equipment repair. GP completions are widely applied in unconsolidated formations to mitigate sand production while maintaining reservoir productivity. However, despite their proven effectiveness in sand control, several post-completion challenges often emerge over time. Common issues such as impaired inflow, screen plugging, and incomplete gravel placement can reduce production rates, accelerate water or gas breakthrough and complicate long-term reservoir management.

In mature reservoirs with complex completions, excessive water production remains a critical challenge. High water cut not only impedes hydrocarbon flow but also increases lifting, separation and disposal cost. In addition, excessive water with high salinity and corrosive gases like CO_2 or H_2S can accelerate corrosion and scaling, leading to frequent equipment failures, downtime, and ultimately reduced net revenue. If left unmanaged, these problems may result in premature field abandonment despite significant remaining hydrocarbon potential. Diagnostic logging is therefore essential to identify water-bearing zones and guide intervention strategies such as chemical water shut-off, mechanical isolation using straddle packers or cement squeezes, and selective recompletion. Pulsed Neutron Logging (PNL) has long been recognized as a valuable diagnostic tool, is capable of monitoring fluid saturations, detect zonal anomalies, and track the movement of water or gas within the reservoir. However, the presence of GP near wellbore introduces signal interferences in PNL measurement and often masking the true formation response and limiting interpretation confidence. This study addresses this limitation by developing an integrated workflow that utilizes both MPDN tool and MCNP nuclear modelling. This approach enables simultaneous evaluation of behind-casing saturation and GP integrity, even in complex sand control completions. With proper linkage to production history, the workflow improves diagnostic certainty, supports informed decision-making for intervention planning, and provides critical input for future infill drilling decisions. The outcome offers operators a practical and cost-effective pathway to optimize production, extend asset life, and maximize hydrocarbon recovery in mature reservoirs.

PNL in GP Completions

The MDPN is an advanced pulsed neutron tool made up of a Deuterium-Tritium (D-T) pulsed neutron generator, four spectroscopic Lanthanum-Bromide (LaBr3) GR detectors, a fast-neutron counter, and processing, control, and telemetry electronics (Fig. 1). The neutron generator emits14 MeV neutrons of into the borehole and surrounding formation. When the neutrons interact with the borehole environment and formation matrix, GR photons are produced and subsequently detected by the MDPN detectors. These GR photons are then converted to light signals within detectors, and the magnitude or amplitude of the light is theoretically proportional to the energy of the GR photons which produced it. Generally, there are four neutron interactions involved - Inelastic Scattering, Elastic Scattering, Capture reaction and Activation events. The MDPN tool records gamma radiation arising from these inelastic and capture interactions, providing valuable insights on both formation and fluids (Loo et al, 2024). The tool operates in three principal data acquisition modes – Sigma, C/O, and Gas mode.

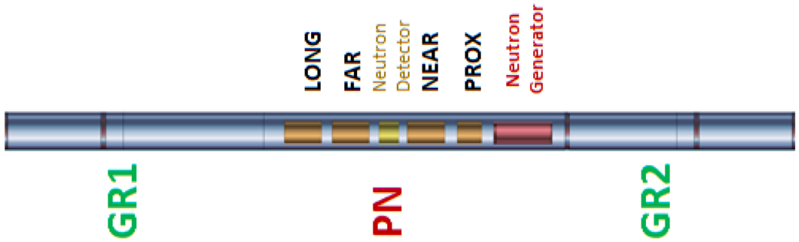

Figure 1—Tool schematic of MDPN tool with top and bottom gamma detector included.

MDPN Gas mode, with a focus on time spectrum measurement, is designed to provide the highest sensitivity to gas within the reservoir while minimizing the influence of formation water salinity. This is achieved through a tailored pulsing sequence and the use of 5 detectors with variable spacing. Since energetic neutron tends to migrate farther when the oil or water in the pore spaces of the formation is replaced with gas, the MDPN Gas mode utilizes signals (sum of neutron counts) from both the closest and furthest detectors to achieve better sensitivity. This results in improved measurement sensitivity and more reliable formation evaluation under challenging downhole conditions. In addition to time spectrum measurement, MDPN tool measures the energies of gamma-rays produced during interactions of "high" energy 14MeV neutrons with atoms in the formation. Carbon and Oxygen are one of those key elements that undergo inelastic interaction. The underlying principle of C/O method is the assumption that carbon atoms were to be found in hydrocarbon molecules (e.g., gas and/or oil, CnH2n+2) and oxygen atoms are primarily associated with formation water (H2O). Thus, depending on the porosity, the C/O ratio would be a direct indicator of formation water saturation. MDPN adopts "Energy Window Solution", where selected spectral energy windows isolate carbon and oxygen signatures. Counts within these windows are summed and ratioed, yielding a quantitative measure of formation saturation independent of water salinity effects.

Evaluating fluid saturation in gravel-pack completions presents several unique challenges. After completion, conventional cased-hole (CH) tools suffer from reduced accuracy in the presence of gravel and shunt tubing structures. Signal attenuation and scattering caused by the gravel material and complex completion geometries including long shunt-tube runs, degrade resolution and complicate interpretation. As described in a multi-zone case, long perforated intervals bridged by shunt-tube systems and variable pack density make it difficult for standard sigma logging and C/O tools to consistently characterize fluid saturations across zones. Researchers have consequently relied on dual-detector capture cross section logs (Sigma logs) and C/O ratio analysis to estimate formation saturation. These are often calibrated using MCNP or field-based analogs to compensate for gravel effects. Time-lapse logging has also been employed to detect changes, although limited spatial resolution and assumptions about uniform gravel distribution remain key limitations. Additionally, production logging tools often fail to resolve inflow contributions because GP alter the near-wellbore flow profile. Collectively, these factors highlight the technical gap in accurately identifying water-bearing intervals and saturation profiles in CH GP wells (Hareezi et al. 2015).

Traditionally, a combined evaluation of pressure survey, material balance, and basic logging data inclusive of the use of radioactive tracer has been adopted to evaluate the quality of GP (Zhang et al. 2017). With the advent of PNL, silicon activation logging is being widely used for GP integrity evaluation. In silicon activation approach, neutron activation is estimated based on the measurement of radiation released by the decay of radioactive nuclei. The nuclei of an atom become excited and unstable that starts to release gamma radiation of certain energies. The unstable radioactive nuclei have its own unique half-life of radiation and for Silicon (Si28) is 134.5s. By accounting for this half-life, logging speed can be optimized to capture the majority of the emitted gamma radiation. Understanding the effects of gravel pack variability on neutron activation response allows uncertainties to be better constrained, thereby improving quantification of behind-casing formation saturation (How et al. 2024). Watson et al. (1991) demonstrated that neutron activation logging using PNL could be best for defining GP quality compared to conventional gamma-gamma density and neutron logging tools, as it is unaffected by changes in borehole environments. In addition to silicon activation logging, other approaches have been utilized to evaluate the gravel packing quality alongside formation saturation. One of the common ones is through analysis of inelastic and capture spectra curves obtained from neutron-GR spectroscopy. This is typically performed by normalizing the near detector inelastic silicon yield to the sum of silicon and iron yields (Amin et al. 2015; How et al. 2024). However, the primary challenge lies in establishing proper elemental offsets or corrections to minimize the influence of poor packing, variations in borehole fluids and completion-related changes on the computed results.

General Sand Screen Completion Design

To mitigate unwanted sand production in producing wells, sand screens are installed and sized gravels are placed in the annular space between a wire-wrapped or slotted screen and the wellbore or casing. Gravels are usually made up of quartz sand or ceramic proppants and the pack is often saturated with brine and is designed to allow hydrocarbon flow while filtering out formation fines. Typical sand screens generally comprise of GP extension, blank pipe and screen (Fig. 2). In a standard WWS type GP, gravel is not usually filled all the way to the packer, but rather just above the screen to ensure complete coverage. Gravel height typically extends 5 to 10 feet above the top of the screen. This is to ensure good sand control with proper packing (no bridging or voids) around the screen. The functions of the different components of sand screen are described as below:

- Sand screen: Primary sand control element allows production fluids to enter the wellbore while preventing sand from entering. Typically made of wire-wrapped, premium mesh, or sintered metal.
- GP extension: A conduit or crossover section between the screen section and the packer or crossover tool. Used to transport gravel down the work string during gravel packing and direct it into the annulus around the screen.
- Blank pipe: A non-perforated section of base pipe with no screen in which it is used to isolate non-producing zones or provide mechanical spacing in the completion.

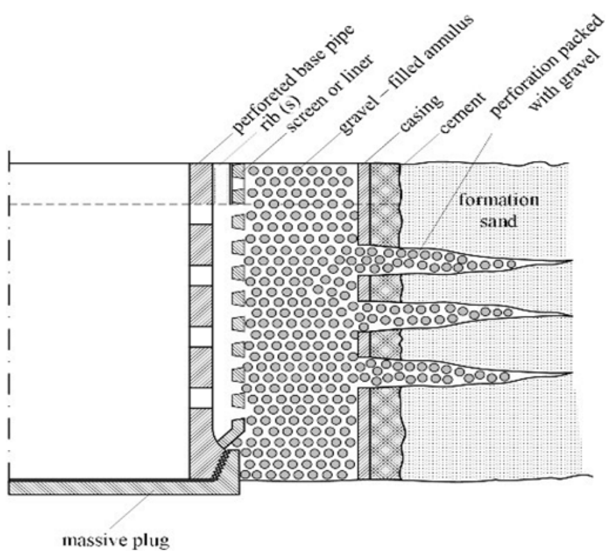

Figure 2—Typical cross section of wire-wrapped screen with its associated components (after Foidaş and Ştefănescu 2017).

Nuclear Modeling Concept

PNL as an effective CH diagnostic method has been widely adopted in industry for formation evaluation. However, in complex gravel-packed completions with tubing, where multiple neutron-interacting interfaces coexist, it complicates the interpretation of PNL responses. The MCNP method, as one of the commonly adopted approaches, provides a stochastic framework to simulate particle transport and interaction physics for such environments, enhancing log interpretation fidelity. The MCNP code, which was developed by Los Alamos National Laboratory, models neutron and photon transport using probabilistic sampling of particle interactions with matter, based on known cross-section libraries. Detectors modeled in MCNP, depending on tool design of each developer, gather simulated spectra or count rates at specified positions, mimicking tool response under various conditions. By understanding the details involved of sand screen components, MCNP simulations allow the establishment of saturation response charts, thereby helping to isolate the effect of each completion element, including variations in gravel porosity, screen material, and fluid saturations behind pipe versus the saturation envelope response from formation.

As shown in Fig. 3, nuclear modeling in GP completion introduces multiple neutron-interacting regions including:

1. Tool – Tubing Interface: Fluid flow area inside the tubing annulus, in which the tubing fluid type can be determined using downlog or uplog pressure.
2. Tubing – Base Pipe/Screen Interface: High-density pipe materials that attenuate neutron and gamma flux. Each of these steel components would have different dimensions (OD & ID), weight, and type along with completion accessories associated with screen. The fluid-filled in annulus between tubing and base pipe could vary depending on the producing or completion fluid.
3. Base Pipe / Screen – Casing Interface: Usually comprised of porous medium (depending on the mesh size used) with a mixture of solid grains and saturating fluids (water, gas, or oil). The GP could have different pack porosity, brine salinity, and elemental material etc.
4. Casing – Formation Interface: The variations in nuclear modeling input falls on reservoir porosity and matrix composition of target reservoir layers. The quality and presence of the cement behind casing also will affect neutron interaction.

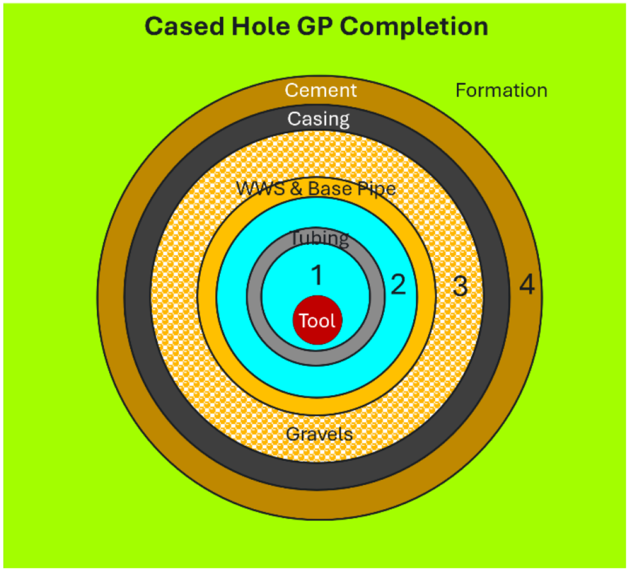

Figure 3—Standard MCNP cross section of CH GP completion with different neutron-interacting regions. 1 = Tool – Tubing interface; 2 = Tubing – Base Pipe / Screen interface; 3 = Base pipe / screen – Casing interface and 4 = Casing – Formation interface.

Each of these regions has its own unique geometries, fluid model, element composition or even reservoir properties, which will then significantly affect the path of how neutrons slow down, thermalize, or even bring captured. This makes nuclear modeling in GP completion a non-standard and tedious process and requires careful handling during simulation stage.

Methodology

Logging Operation

In oil and gas saturation analysis for GP completion, typically under 24 hours operation, minimum 2 run will be required.

- Run 1: C/O logging with logging speed 3fpm – Typically to be done first as silicon activation works best during the first run without any activation to the formation. C/O up to 4 passes can be run if the statistical variation of the data (standard deviation) is high especially in GP section.
- Run 2: Gas mode logging with logging speed of 8fpm or Sigma logging with logging speed of 15fpm. Typically require main & repeat pass for repeatability check. In the case of CH gas quantification analysis fresh water environment, Gas mode is preferred due to its salinity independent nature.

Silicon Activation

For MDPN tool, silicon activation logging can be performed in every logging mode with two GR tools. One is located above the detector array (GR1 or top GR) and second one is located about 10ft below neutron generator (GR2 or bottom GR). Top GR1 provides total counts of naturally occurring gammas from radioactive elements as U, Th and K, while bottom GR2 provides total counts of naturally occurring gammas including gammas from activated silica (Fig. 4). The high-energy neutrons are pulsed and activated the Silicon (Si) elements in GP. Internal studies showed that silicon activation measurements obtained from C/O mode with logging speed of 3fpm is preferred as the largest number of gammas from activated Silicon can be obtained.

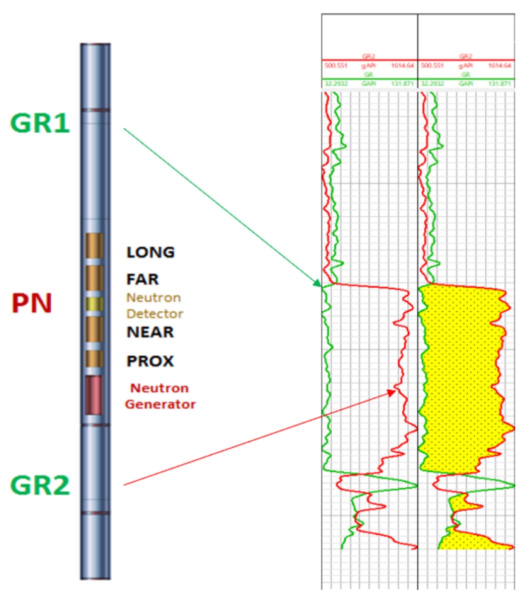

Figure 4—Concept of background GR versus total GR for silicon activation logging. Log track 1 represents the natural GR obtained from top GR sensor (green curves) and the total GR including the activated GR retrieved from bottom GR sensor (red curves). Yellow shading in track 2 represents the activated GR which is the delta between natural GR and total GR.

Nowadays, to save operation time, silica activation measurements can be extracted concurrently during the 1st logging run. 1st logging pass is chosen mainly because the formation has not been activated after shut-in and with minitron turned off at first. The background GR from the formation can be obtained as the baseline. Once the minitron is turned on, the induced GR can now be captured in the bottom GR sensor when passing through the activated GP completion. Si elements within the gravel packing material are detected and the amount of Si present indicates the GP quality. To quantify the GP packing efficiency, analyst can first quantify the difference between natural GR versus the total GR from 1st logging pass data. Based on the curve response, proper scaling of delta GR of 100% packed section versus the section with 0% packing shall give indication of overall GP quality. From here, the overlay of measured top GR and bottom GR shall also suggest the top depth of GP. However, it is worth noting that outliers relating to completion accessories shall be removed prior to computation of GP packing percentage.

Pre-Job Nuclear Modeling

In this study, expected values of Gas mode burst and capture ratios, as well as C/O ratios were characterized using MCNP probabilistic simulation modelling to support quantitative saturation analysis. For this specific application, GP and screen fluids were modeled independently of formation and tubing / casing fluid. GP characterization parameters included the known dimensions of screens and base pipe (OD and ID), the composition and type of the gravels, and the porosity of the GP surrounding the tool. Meanwhile, formation model parameters focus primarily on the number of casing and tubing steels, their dimensions, annulus fluid type, and reservoir porosity. The MCNP modelling workflow generated response equations for each MDPN detector under different input porosities for both GP and formation scenarios. From these equations, expected saturation envelopes were predicted. The critical challenge in behind-casing saturation evaluation lies in the ability to distinguish GP-related responses from those originating in the reservoir, which requires careful analysis of detector sensitivity. For instance, in case study 1, a total of 8 MCNP models which comprised of 6 formation models and 2 GP models were generated to account for variations in tubing-screen-casing fluid type.

Gas Ratios and C/O Dual-Saturation Analysis in GP Completions

For gas saturation analysis, the MDPN tool primarily uses the burst or capture Proximal-Long ratio measurements. Gas-filled zones typically result in higher neutron counts at the distal detectors; consequently, a lower burst Proximal-Long ratio typically indicates the presence of gas. By analyzing this ratio across different detector spacings, the MDPN tool can resolve both the vertical distribution and relative concentration of gas within and beyond the GP interval. Moreover, the system allows flexible interpretation by selecting desired detector ratio combinations from the burst or capture time spectrum. For instance, by giving more emphasis to capture time spectrum, it can help detect gas deeper in the formation; while burst time spectrum are more effective for detecting gas near the casing or within the GP. This algorithm enables a more robust assessment of gas saturation across varying DOIs. On the other hand, MDPN tool adopted a different approach for computing C/O saturation, which is by combining two response equations for hydrocarbon saturation analysis. Instead of analyzing C/O ratio versus porosity and Ca/Si (Calcium/Silicon ratio) versus porosity independently, this method integrates both relationships by constructing C/O-Ca/Si ratio cross plot with porosity polygons as third dimension on C/O versus Ca/Si cross-plot (Fig. 5). C/O sensitivity analysis was performed using two different detector weightage mixtures at one time. The intent of this C/O dual-saturation workflow is to distinguish oil signatures at different DOI. For example, assigning 100% weight to the Proximal detector combined with partial weighting from the Near detector enhances sensitivity to fluids in the GP zone, whereas greater emphasis on the distal detectors improves characterization of saturation within the reservoir. This combined approach provides a more comprehensive picture of oil and gas distribution across multiple radial zones around the wellbore.

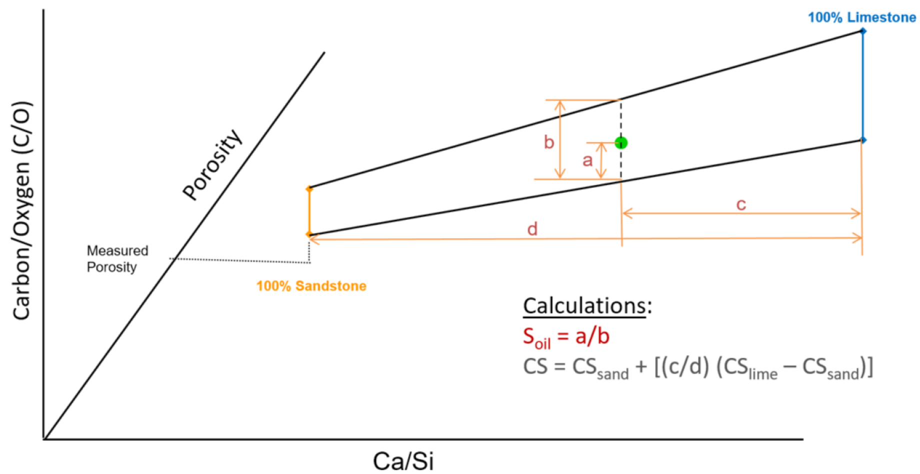

Figure 5—MDPN C/O MCNP cross plot of C/O ratio versus Ca/Si ratio. C/O saturation model, which is represent in 4-sided polygon, is a function of porosity and lithology.

However, in sand screen completions, GP models with varying characteristics such as gravel porosity, screen configuration, screen dimension, weight etc. exhibit different MCNP response compared to conventional completion without GP. It is observed that after model normalization, the GP model tends to fall ahead of the formation model (Fig. 6). For accurate formation saturation quantification, the analyst's goal is to normalize the data to locations which are unaffected by GP model, while accounting for the porosity of both GP (about 37%) and the formation (about 24%) (Fig. 7). GP porosity is not a fixed value and can vary significantly depending on several factors such as confining pressure, mean grain size, sorting, packing efficiency and the type of gravel material used. For example, quartz sand packs typically have lower porosity compared to ceramic proppant. Under a confining pressure of 10MPa, quartz sand GP generally exhibits porosity in the range of 40-55%; whereas ceramic-type proppants range from 45-55%. Importantly, the influence of GP or borehole environmental effects diminishes with increasing detector spacing from the neutron generator, as distal detectors are less sensitive to near-wellbore heterogeneities.

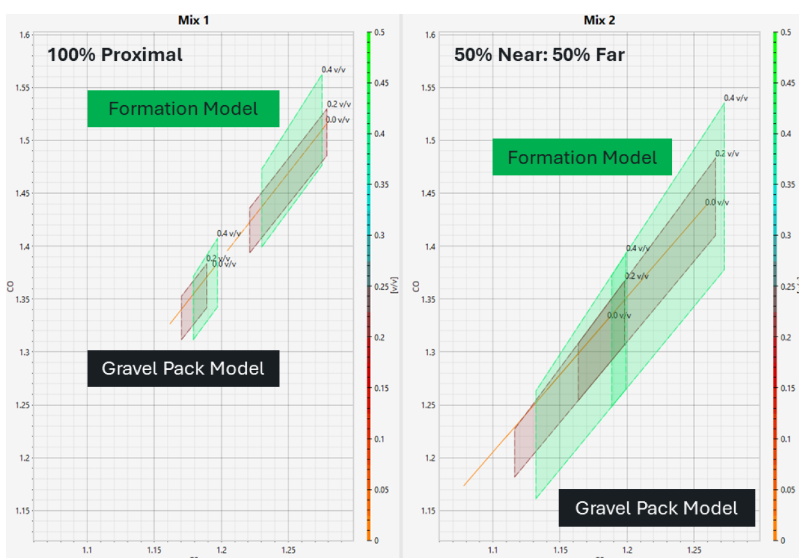

Figure 6—Comparison of GP model and formation model in different C/O detector mixture weightage of 100% Proximal (Mix 1) versus 50% Near and 50% Far (Mix 2) after model normalization.

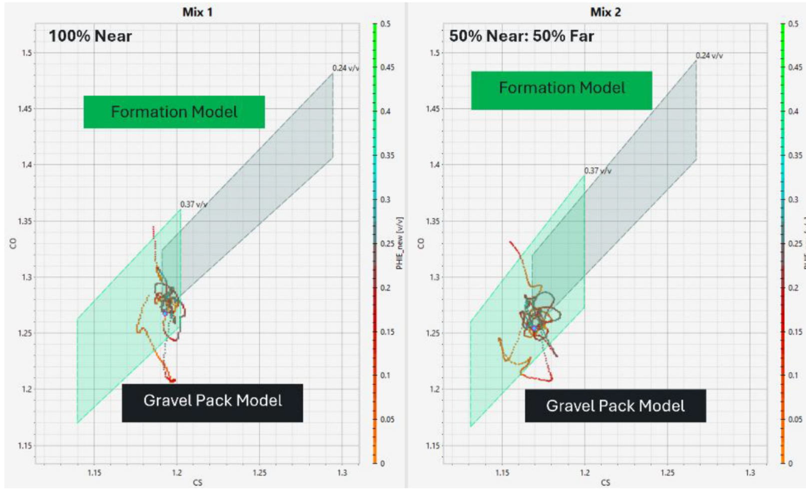

Figure 7—Comparison of GP model and formation model in different C/O detector mixture weightage of 100% Proximal (Mix 1) versus 50% Near and 50% Far (Mix 2) with specified GP porosity of 37% and reservoir porosity of 24%. Data clusters are normed to the water zone with the least effect from the GP.

Results & Discussion

Case Study Example 1: Clastic formation with 4-in WWS in 7-in casing and gas-filled 2.375-in tubing

This case study is conducted as part of the data acquisition for derisking studies on Field X. MDPN was logged inside 2.375-in tubing with WWS using Gas mode and C/O for contact and saturation monitoring. The saturation analysis complicates when most of the targeted sands are within the dual tubing or single tubing with GP, together with changes in fluid types present in different tubing-screen-casing interface. Pressure gradient analysis shows that gas is filling inside the tubing, and this may increase the uncertainty in computed saturation since the saturation envelope with gas in tubing especially for gas identification will be much less compared to logging with fluid inside tubing. As this is not expected during the initial pre-job planning, a different MCNP model with gas-filled tubing was acquired and used for analysis.

The main PNL log responses and saturation results are presented in Fig. 8. Based on the Open Hole (OH) analysis, gas was only present in the topmost sand below the packer of dual tubing section till x70ft MD with oil evident in most of the sand layers until the bottommost section. Recent PNL results showed that within the OH gas-proned zone, there could be potential gas trapped below the packer. This is hypothesized from the observations that the Proximal-Long burst ratio (VB15C) is reading extremely low with not much contrast observed even in sand vs shale section. At the same time, the drastic increase of C/O ratios near the EOT of short string and above the production packer also suggests the possibility of oil trapped in borehole annulus. All these variations in annulus fluid types prompted the need of careful zonation during processing. To minimize the effect from borehole fluid, capture gas saturation is calculated using Proximal-Long capture ratios (VC15C). With current normalization at shale, formation gas is evident with possible gas-liquid contact (GLC) moving down to X40ft MD. Formation sigma reading below 15 capture unit also confirms the presence of formation gas. Below X40ft MD, mixture of dominant oil with some gases are evident based on MDPN result. This is tally with the production result of a slightly downdip offset well, where perforation was done and still producing good oil rate with low Gas-Oil ratio.

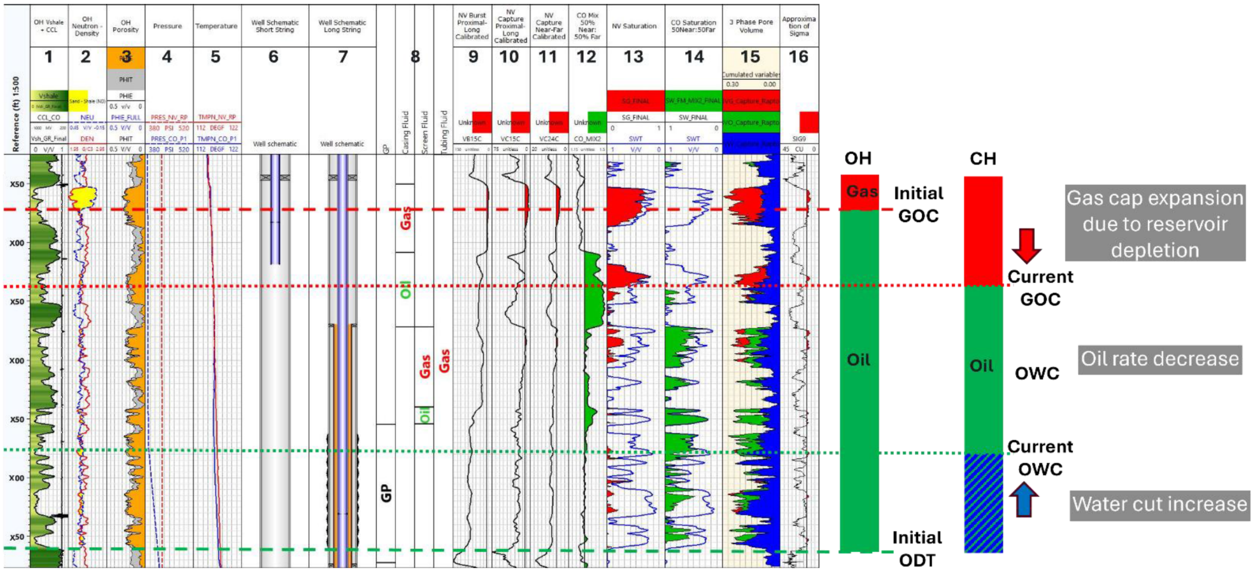

Figure 8—MDPN Gas and C/O saturation results with possible contact movement and reservoir findings. Track 1 – OH volume of shale and casing collar locator; 2 – OH density-neutron; 3 – OH porosities; 4 – Pressure; 5 – Temperature; 6 & 7 – Well schematic of short and long string; 8 – GP zone and fluid type of each neutron interacting region; 9 – Proximal-Long burst ratio from Gas mode; 10 - Proximal-Long capture ratio from Gas mode; 11 – Near-Far capture ratio from Gas mode; 12 – C/O mixed ratio from 50% Near and 50% Far; 13 – MDPN gas saturation; 14 – MDPN C/O oil saturation; 15 – MDPN 3 phase pore fluid volume; 16 – Approximation of sigma

Moving on to the 4-in WWS section, the drastic drop in VB15C again triggered the possibility of having gas but this time could be within the annulus space below the GP extension and unperforated base pipe (shaded red, as seen in Fig. 9). It is worth noting that gravels may not be fully topped up till the packer as what normally presumed and in fact, a detailed illustration of the GP together with top of gravel identified from silicon activation analysis are of inestimable help. Fig. 10 illustrates the silicon activation analysis performed with result indicates that top of gravel is located just a few feet above the current perforation with GP percent of about 70% for upper sand formation. In terms of packing quality, most of the gravels seem to be still intact with no voids or changes in packing efficiency observed. The increase in Proximal Silica count also proves the presence of gravels within the screen. The poor GP integrity may cause oil trapped inside the gravels, giving optimistic saturation results. As compared to upper sand formation, the lower sand formation with another 4-in WWS shows a slightly better packing, but with more variation in packing efficiency, ranging from 60 – 80%. Extra precaution required during formation saturation computation as the change in tubing fluid from gas to oil would increase the risk of having oil trapped within the gravel voids, giving overestimated final C/O saturation. Traditionally, service providers had relied primarily on pre-job modelling carried out ahead of GP and the pressure response from surface or wash-pipe gauges to confirm pack efficiency in the past. However, with the introduction of silicon activation logging, complemented by inelastic Silica count measurements from multiple detectors, analysts now could quantitatively assess the packing efficiency with greater confidence. This advancement reduces dependency on indirect surface measurements and minimizes the need for costly well interventions, thereby providing a more accurate and cost-effective means of evaluating GP integrity.

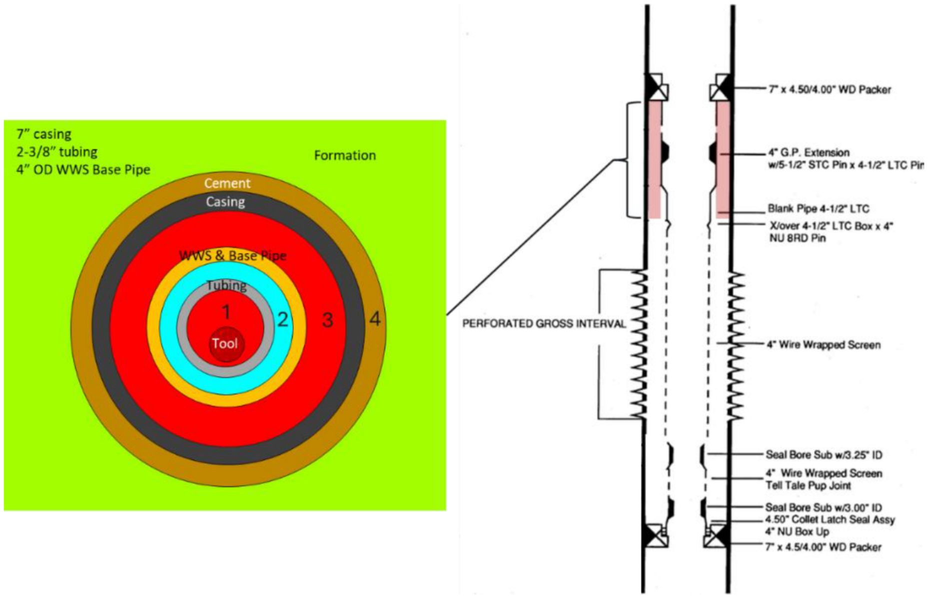

Figure 9—MCNP cross section (left) along with GP completion schematic for upper formation (right), illustrating the possible gas trapped beneath the WD packer inside the behind screen (base pipe)-casing interface (label 3 in cross section).

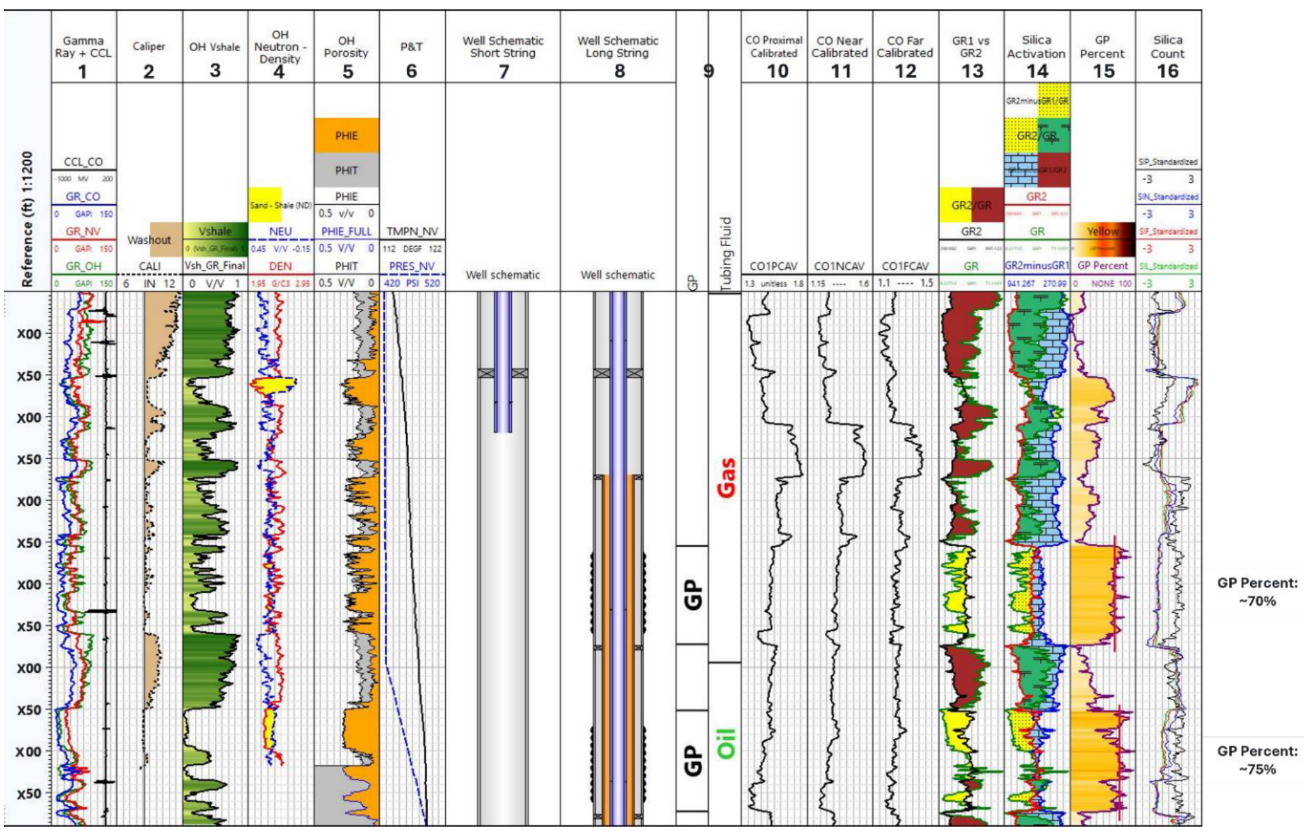

Figure 10—Silicon activation analysis result with estimated GP percent for each WWS section. Track 1 – OH GR and casing collar locator; 2 – OH caliper; 3 – OH volume of shale; 4 – OH density-neutron ; 5 – OH porosities; 6 – Pressure and temperature; 7 – Well schematic of short string; 8 – Well schematic of long string; 9 – GP zone and tubing fluid type; 10 – Average C/O ratio from Proximal detector; 11 – Average C/O ratio from Near detector; 12 – Average C/O ratio from Far detector; 13 – Overlay of top GR versus bottom GR; 14 – Silica activation analysis for lithology identification using top, bottom and delta GR measurements; 15 – Estimated GP packing percentage; 16 – Inelastic Silica count from each MDPN detector.

In addition, based on Fig. 8, the drop in C/O ratio and increase in VB15C and VC15C within the gravel-filled WWS section indicate that water could be present within the annulus-behind screen-casing interface.

With normalization on lowest C/O mix ratio recorded at bottommost section, decrease of oil saturation down to So of 30% could be observed, indicating the sign of water up. To better understand the uncertainty, different detector mixture weightages are adopted with Mix 1 using 100% Near and Mix 2 using 50% Near: 50 Far. Saturation analysis from both detector mixtures illustrated the similar findings in terms of overall contact movement with only variations seen in computed saturation. As GP is present near wellbore, it is more logical to have heavier weightage on further detectors for formation saturation evaluation. The availability of both GP and formation nuclear MCNP models helps log analysts to better understand how heavy the neutron response is being affected by the near wellbore sand screen and what could be the expected uncertainty when computing reliable saturation estimation. From reservoir evaluation perspective, the MDPN results indicate that the gas could have expanded due to reservoir depletion. Decrease in current saturation matches with the current observation of oil rate decreases. Current oil-water contact also moves up, tally with evidence of water cut increase in this well. As part of the mitigation plan to isolate the excessive water at bottommost section, depending on the type of hydrocarbon targeting and complexity of the well intervention job, user may shut off chemically through pumping via capillary unit and perform thru-tubing perforation at upper section for additional gas.

Case Study Example 2: Clastic formation with 5.5-in WWS in 9.625-in casing and water-filled 3.5-in tubing

MDPN was deployed inside 3.5-in tubing with WWS using Gas mode and C/O for contact and saturation monitoring. Fig. 11 shows the gas and oil saturation results for this well. OH analysis indicated that the original GLC could be located approximately at x75ft MD with oil present at bottom section, separated by shale buffer. Pressure gradient analysis shows that water is filling inside the tubing. However, the extremely low, characterless burst Proximal-Long ratios suggests that gas could be potentially trapped below the packer within the base pipe – casing annulus region at Zone_02. With current normalization at shale section below the OH gas-proned zone and appropriate MCNP model, some significant formation gas responses are evident at Zone_02 at least till the depth of original GLC. The decrease in C/O ratio also supports the hypothesis of gas-proned sand in Zone_02. From saturation computed from VC15C, which tends to have deeper DOI, provides a better estimation on final gas saturation. Gas saturation computed from VB15C may be more affected by the presence of GP. Near the bottom of Zone_02, the subtle increase in raw C/O Near and Far ratios suggest potentially some remaining oil in unperforated section. Meanwhile, for Zone_03 and Zone_04, the increase in VB15C indicates fluid responses. This suggests that Zone_03 and Zone_04 should be mostly water and this observation matches 90% water cut evident in most recent well test. In addition, based on the PVT correlation, the estimated bubble point pressure for Zone_02 sand is ~9XX psia and the current reservoir suggests that it is now a saturated reservoir with estimated pressure of approximately ~5XX psia. Interestingly near the bottom of Zone_04 at X35ft, the silicon activation GR overlay method suspects that there could be deterioration in GP packing quality, which leads to anomalous high C/O saturation. A better GP packing quality is evident in Zone_04.

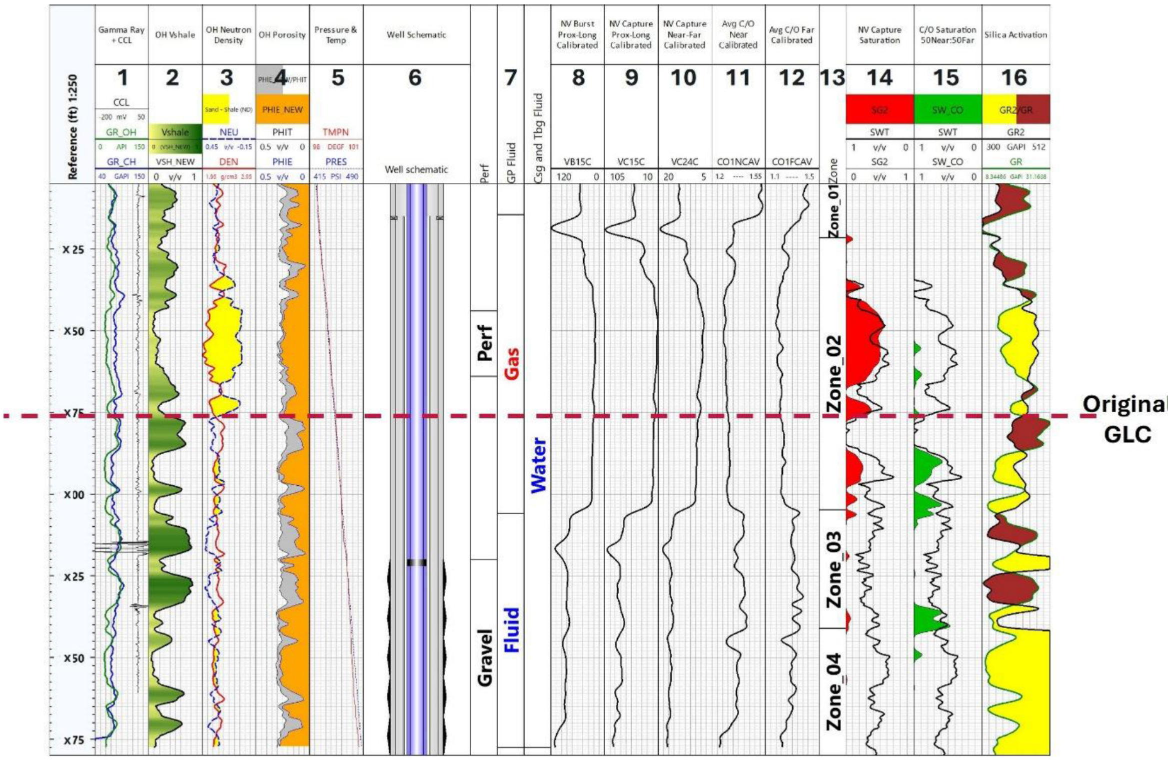

Figure 11—MPDN Gas and C/O saturation results with its associated zonation based on changes in fluid distribution across sand control completion. Track 1 – OH and CH GR with casing collar locator; 2 – OH volume of shale; 3 - OH density-neutron; 4 – OH porosities; 5 – Pressure and temperature; 6 – Well schematic; 7 – Perforation and GP zone with fluid type of each neutron interacting region; 8 – Proximal-Long burst ratio from Gas mode; 9- Proximal-Long capture ratio from Gas mode; 10 – Near-Far capture ratio from Gas mode; 11– Average C/O ratio from Near detector; 12 – Average C/O ratio from Far detector; 13 – Zonation based on changes of fluid-filled in tubing-screen-casing interfaces; 14 – MDPN gas saturation; 15 – MDPN C/O oil saturation; 16 – Overlay of top GR versus bottom GR.

As part of the post-logging production optimization plan, perforation was done on Zone_02, targeting the gas sand. Production suggested good gas rates on the first day. However, during the second day of production, the well starts to produce mostly water with small amounts of oil and gas. This suggests that the shale buffer at bottom of Zone_02 is not strong enough to provide the isolation and Zone_02 and 03 sand could be connected. Once water starts to build up, it could trigger liquid loading and formation gas may not have sufficient energy to lift to surface after prolong period. Nevertheless, the additional gas was used to supply the usage for nearby facilities intermittently when required.

Based on the observations from the above case studies, some of the challenges faced during nuclear modeling include:

1. Uncertainty in sand screen info – During modeling stage, analyst would require details on sand screen type used and the porosity of the gravels. Most of the customers do not have details on the installed sand screen, especially for those sand screens that were installed years ago with no proper info storage system in place. Assumptions need to be made sometimes.
2. Uncertainty in fluid type – With focus on CH GP, it is well noted that apart from tubing annulus fluid that is being defined by tubing pressure measurement, there are additional fluid flow area between tubing - screen and screen - casing interfaces, in which fluid-filled type could be vary across the logging depth. Some initial assumptions on fluid type would need to be made prior to MCNP modeling based on observation from raw data and thus far, it has been providing reasonable analysis on the case studies.
3. Complexity in modeling process - Longer period of simulation due to increase in number of steels and neutron interacting interfaces.

Conclusion and Way Forward

This study demonstrated that PNL surveillance using MDPN Gas and C/O modes is an effective approach for behind-casing formation evaluation, even in complex sand screen completions. By explicitly accounting for tubing–screen–casing interfaces and near-wellbore complexities, the workflow enables clear separation of true reservoir responses from screen-dominated near-wellbore effects—historically one of the major limitations in PNL interpretation.

The integration of MDPN multi-detector DOI capabilities with physics-based nuclear modeling provides a cost-effective alternative to intervention-based diagnostics. Advancements in quantitative gas saturation interpretation and C/O dual-saturation techniques, supported by GP –specific MCNP modeling, allow true reservoir signals to be distinguished with higher confidence. Gas differentiation using time-spectrum ratio analysis successfully identified responses from trapped annular gas versus true reservoir fluids, thereby preventing misinterpretation of gas-bearing intervals. Similarly, advanced C/O saturation analysis enabled early diagnosis of water breakthrough in perforated zones, consistent with high water cut trends observed in production history. The field results confirm the robustness of this integrated approach and improve the accuracy of determining current fluid contacts and saturation in mature assets. Beyond formation evaluation, gravel pack integrity assessment using silicon activation proved a valuable addition. This technique enabled identification of packing efficiency and deterioration within the WWS interval, providing operators with critical insights into sand control effectiveness and supporting timely intervention planning. For brownfield assets where sand and water management are essential to prolong field life, the integrated MDPN–MCNP methodology offers a practical solution to acquire high-value reservoir and completion integrity information without costly workovers.

In conclusion, MDPN surveillance supported by nuclear modeling should be regarded not merely as a diagnostic tool but as a cornerstone of reservoir management in complex sand control completions. Wider adoption of this integrated workflow across mature fields will enhance production optimization, strengthen asset integrity, and support sustainable field development. Future work is recommended to investigate the impact of different screen types and materials, such as gadolinium-treated sand screens, on PNL responses to ensure proper quantitative corrections are applied in formation saturation calculations.

Nomenclature

C/O	Carbon / Oxygen
Ca/Si	Calcium / Silicon
CH	Cased-Hole
DOI	Depth of Investigation
GP	Gravel Pack
GR	Gamma Ray
MCNP	Monte Carlo N-Particle
MDPN	Multi Detector Pulsed Neutron
OH	Open Hole
PNL	Pulsed Neutron Logging
WWS	Wire Wrapped Screen

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